EC200U&EG91xU Series

QuecOpen(SDK) Jamming Detection Development Guide

LTE Standard Module Series

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About the Document

Revision History

Version	Date	Author	Description
-	2023-08-08	Farris FENG	Creation of the document
1.0	2023-11-15	Farris FENG	First official release

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1 Introduction

Quectel EC200U series and EG91xU family modules support QuecOpen® solution. QuecOpen® is an embedded development platform based on RTOS. It is intended to simplify the design and development of IoT applications. For more information on QuecOpen®, see ***document [1]***.

This document is applicable to QuecOpen® solution based on CSDK build environment. It introduces the development guide of jamming detection, including the jamming detection APIs and development examples on application layer of Quectel EC200U series and EG91xU family modules.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Module
-	EC200U Series
EG91xU	EG912U-GL
	EG915U Series

2 Jamming Detection API

2.1. Header File

ql_api_nw.h, the header file of jamming detection API, is in the *\components\quectel\ql_open\ql_api\include* directory of CSDK. Unless otherwise specified, all the header files mentioned in this document are in this directory.

2.2. API Overview

Table 2: API Overview

Function	Description
ql_nw_set_jamdet_mode()	Enables or disables the jamming detection feature.
ql_nw_get_jamdet_mode()	Gets the enabled/disabled status of the jamming detection feature.
ql_nw_get_jamdet_status()	Gets the current detected jamming status.
ql_nw_set_jamdet_config()	Sets the parameters of jamming detection feature.
ql_nw_get_jamdet_config()	Gets the current parameter configuration of jamming detection.
ql_nw_set_jamdetex_config()	Sets the extended parameters of jamming detection feature.
ql_nw_get_jamdetex_config()	Gets the current extended parameter configuration of jamming detection.

2.3. API Description

2.3.1. ql_nw_set_jamdet_mode

This function enables or disables the jamming detection feature.

- **Prototype**

```
ql_nw_errcode_e ql_nw_set_jamdet_mode(uint8_t nSim, uint8_t enable)
```

- **Parameter**

nSim:
[In] The (U)SIM card to be used. If the module only supports one (U)SIM interface, this parameter can be set to *0*.
  0 (U)SIM card 1
  1 (U)SIM card 2

enable:
[In] Enable or disable the jamming detection feature.
  0 Disable
  1 Enable

- **Return Value**

See ***Chapter 2.3.1.1*** for details.

2.3.1.1. ql_nw_errcode_e

The error code enumeration for the jamming detection API:

```
typedef enum
{
    QL_NW_SUCCESS                    = 0,
    QL_NW_EXECUTE_ERR                = 1 | (QL_COMPONENT_NETWORK << 16),
    QL_NW_MEM_ADDR_NULL_ERR,
    QL_NW_INVALID_PARAM_ERR,
    QL_NW_CFW_CFUN_GET_ERR,
    QL_NW_CFUN_DISABLE_ERR           = 5 | (QL_COMPONENT_NETWORK << 16),
    QL_NW_CFW_NW_STATUS_GET_ERR,
    QL_NW_NOT_SEARCHING_ERR,
    QL_NW_NOT_REGISTERED_ERR,
    QL_NW_CFW_GPRS_STATUS_GET_ERR,
    QL_GPRS_NOT_SEARCHING_ERR        = 10 | (QL_COMPONENT_NETWORK << 16),
```

```
        QL_GPRS_NOT_REGISTERED_ERR,
        QL_NW_CFW_NW_QUAL_GET_ERR,
        QL_NW_CFW_OPER_ID_GET_ERR,
        QL_NW_CFW_OPER_NAME_GET_ERR,
        QL_NW_CFW_OPER_SET_ERR             = 15 | (QL_COMPONENT_NETWORK << 16),
        QL_NW_SIM_ERR,
        QL_NW_NO_MEM_ERR,
        QL_NW_SEMAPHORE_CREATE_ERR,
        QL_NW_SEMAPHORE_TIMEOUT_ERR,
        QL_NW_NITZ_NOT_UPDATE_ERR        = 20 | (QL_COMPONENT_NETWORK << 16),
        QL_NW_CFW_EMOD_START_ERR,
        QL_NW_OPERATOR_NOT_ALLOWED,
        QL_NW_CFW_RRCRel_SET_ERR,
        QL_NW_BLACK_CELL_FULL_ERR,
        QL_NW_PLMNFORMAT_CONVERT_UNFOUND = 25 | (QL_COMPONENT_NETWORK << 16),
        QL_NW_CFW_NW_INFO_GET_ERR,
}ql_nw_errcode_e;
```

- **Member**

Member	Description
QL_NW_SUCCESS	Successful execution.
QL_NW_EXECUTE_ERR	Failed execution.
QL_NW_MEM_ADDR_NULL_ERR	The API parameter is null.
QL_NW_INVALID_PARAM_ERR	Invalid API parameter(s).
QL_NW_CFW_CFUN_GET_ERR	Failed to get functional mode.
QL_NW_CFUN_DISABLE_ERR	Failed to disable the functional mode.
QL_NW_CFW_NW_STATUS_GET_ERR	Failed to get the network status.
QL_NW_NOT_SEARCHING_ERR	The operator to be registered is not found.
QL_NW_NOT_REGISTERED_ERR	The network is not registered.
QL_NW_CFW_GPRS_STATUS_GET_ERR	Failed to get PS domain network status.
QL_GPRS_NOT_SEARCHING_ERR	No operator to be registered for PS domain is found.
QL_GPRS_NOT_REGISTERED_ERR	Not registered on a PS domain network.
QL_NW_CFW_NW_QUAL_GET_ERR	Failed to get signal quality.

QL_NW_CFW_OPER_ID_GET_ERR	Failed to get operator ID.
QL_NW_CFW_OPER_NAME_GET_ERR	Failed to get operator name.
QL_NW_CFW_OPER_SET_ERR	Failed to set operator.
QL_NW_SIM_ERR	(U)SIM card related error.
QL_NW_NO_MEM_ERR	Failed to allocate memory.
QL_NW_SEMAPHORE_CREATE_ERR	Failed to create semaphore.
QL_NW_SEMAPHORE_TIMEOUT_ERR	Waiting for semaphore timeout.
QL_NW_NITZ_NOT_UPDATE_ERR	Unsynchronized network time.
QL_NW_CFW_EMOD_START_ERR	Failed to enter engineering mode, resulting in cell information unavailable.
QL_NW_OPERATOR_NOT_ALLOWED	Operation is not allowed.
QL_NW_CFW_RRCRel_SET_ERR	Failed to set RRC fast release.
QL_NW_BLACK_CELL_FULL_ERR	Blacklisted cell is full.
QL_NW_PLMNFORMAT_CONVERT_UNFOUND	PLMN format conversion target not found.
QL_NW_CFW_NW_INFO_GET_ERR	Failed to get network information.

2.3.2. ql_nw_get_jamdet_mode

This function gets the enabled/disabled status of the jamming detection feature.

- **Prototype**

```
ql_nw_errcode_e ql_nw_get_jamdet_mode(uint8_t nSim, uint8_t *enable)
```

- **Parameter**

nSim:
[In] The (U)SIM card to be used. If the module only supports one (U)SIM interface, this parameter can be set to *0*.
0 (U)SIM card 1
1 (U)SIM card 2

enable:
[Out] The status of the jamming detection feature.
0 Disabled
1 Enabled

- **Return Value**

See ***Chapter 2.3.1.1*** for details.

2.3.3. ql_nw_get_jamdet_status

This function gets the current detected jamming status.

- **Prototype**

```
ql_nw_errcode_e ql_nw_get_jamdet_status(uint8_t nSim, ql_nw_jamdet_state_e *status)
```

- **Parameter**

nSim:
[In] The (U)SIM card to be used. If the module only supports one (U)SIM interface, this parameter can be set to *0*.
0 (U)SIM card 1
1 (U)SIM card 2

status:
[Out] The current detected jamming status. See ***Chapter 2.3.3.1*** for details.
0 No jamming
1 Jamming detected

- **Return Value**

See ***Chapter 2.3.1.1*** for details.

2.3.3.1. ql_nw_jamdet_state_e

The enumeration of the current detected jamming status:

```
typedef enum
{
    QL_NO_JAMMING = 0,
    QL_JAMMED,
}ql_nw_jamdet_state_e
```

- **Member**

Member	Description
QL_NO_JAMMING	No jamming.
QL_JAMMED	Jamming detected.

2.3.4. ql_nw_set_jamdet_config

This function sets the parameters of jamming detection feature.

- **Prototype**

```
ql_nw_errcode_e ql_nw_set_jamdet_config(uint8_t nSim, quec_jamdet_param_set_s *config)
```

- **Parameter**

nSim:
[In] The (U)SIM card to be used. If the module only supports one (U)SIM interface, this parameter can be set to *0*.
0 (U)SIM card 1
1 (U)SIM card 2

config:
[In] The structure of the jamming detection parameter configuration. The length is equal to the number of members in the *quectel_jdcfg_e_t* enumeration. See ***Chapter 2.3.4.1*** for details.

- **Return Value**

See ***Chapter 2.3.1.1*** for details.

2.3.4.1. quec_jamdet_param_set_s

The structure of the jamming detection parameter configuration:

```
typedef struct
{
    quectel_jdcfg_e_t  qjdcfg_enum;
    int                value;
}quec_jamdet_param_set_s
```

- **Parameter**

Type	Parameter	Description
quectel_jdcfg_e_t	*qjdcfg_enum*	Parameter type. See ***Chapter 2.3.4.2*** for details.
int	*value*	Parameter value.

2.3.4.2. quectel_jdcfg_e_t

The enumeration of parameter types:

```
typedef enum
{
    JDCFG_ENUM_NONE = -1,
    JDENABLE  = 0,
    JDNSIM,
    JDURC,
    JDPERIOD,
    JDMINCH,
    JDSINR,
    JDRSSI_GSM,
    JDRSRP,
    JDRSRQ,
    JDRSSI,
    JDCFG_ENUM_MAX
 }quectel_jdcfg_e_t;
```

- **Member**

Member	Description
JDCFG_ENUM_NONE	Invalid value.
JDENABLE	Enabled/disabled status.
JDNSIM	(U)SIM card number.
JDURC	URC enabled/disabled status (Not supported currently)
JDPERIOD	URC reporting period (Not supported currently)
JDMINCH	Channel number threshold for determining whether there is a jamming in the GSM cell. Range: 0–254. Default: 5.
JDSINR	SINR threshold for determining whether there is a jamming in the GSM cell. Range: 0–63. Default: 3.

JDRSSI_GSM	RSSI threshold for determining whether there is a jamming in the GSM cell. Range: -110 to -50. Default: -80.
JDRSRP	RSRP threshold for determining whether there is a jamming in the LTE cell. Range: -140 to -44. Default: -105.
JDRSRQ	RSRQ threshold for determining whether there is a jamming in the LTE cell. Range: -19 to -3. Default: -15.
JDRSSI	RSSI threshold for determining whether there is a jamming in the LTE cell. Range: -120 to -20. Default: -50.
JDCFG_ENUM_MAX	Maximum value of the enumeration type.

NOTE

JDRSSI_GSM applies only to the modules that use R03 firmware versions in QuecOpen® solution. R03 firmware refers to firmware whose version number contains a segment “R03”.

2.3.5. ql_nw_get_jamdet_config

This function gets the current parameter configuration of jamming detection.

- **Prototype**

```
ql_nw_errcode_e ql_nw_get_jamdet_config(uint8_t nSim,quec_jamdet_param_set_s *config)
```

- **Parameter**

nSim:

[In] The (U)SIM card to be used. If the module only supports one (U)SIM interface, this parameter can be set to *0*.

0 (U)SIM card 1

1 (U)SIM card 2

config:

[Out] The structure of the jamming detection parameter configuration. The length is equal to the number of members in the *quectel_jdcfg_e_t* enumeration. See ***Chapter 2.3.4.1*** for details.

- **Return Value**

See ***Chapter 2.3.1.1*** for details.

2.3.6. ql_nw_set_jamdetex_config

This function sets the extended parameters of jamming detection feature.

- **Prototype**

```
ql_nw_errcode_e ql_nw_set_jamdetex_config(uint8_t nSim, quec_jamdetex_param_set_s *config)
```

- **Parameter**

nSim:
[In] The (U)SIM card to be used. If the module only supports one (U)SIM interface, this parameter can be set to *0*.
0 (U)SIM card 1
1 (U)SIM card 2

config:
[Out] The structure of jamming detection extended parameter configuration. See ***Chapter 2.3.6.1*** for details.

- **Return Value**

See ***Chapter 2.3.1.1*** for details.

NOTE

This function applies only to the modules that use R03 firmware versions in QuecOpen® solution. R03 firmware refers to firmware whose version number contains a segment "R03".

2.3.6.1. quec_jamdetex_param_set_s

The structure of jamming detection extended parameter configuration:

```
typedef struct
{
    uint8_t shakeperiod;
}quec_jamdetex_param_set_s;
```

- **Parameter**

Type	Parameter	Description
uint8_t	*shakeperiod*	Anti-shake duration. Range: 1–10. Unit: second. Default: 5.

2.3.7. ql_nw_get_jamdetex_config

This function gets the current extended parameter configuration of jamming detection.

- **Prototype**

```
ql_nw_errcode_e ql_nw_get_jamdetex_config(uint8_t nSim,quec_jamdetex_param_set_s *config)
```

- **Parameter**

nSim:
[In] The (U)SIM card to be used. If the module only supports one (U)SIM interface, this parameter can be set to *0*.
0 (U)SIM card 1
1 (U)SIM card 2

config:
[Out] The structure of jamming detection extended parameter configuration. See ***Chapter 2.3.6.1*** for details.

- **Return Value**

See ***Chapter 2.3.1.1*** for details.

3 Development Example

This chapter demonstrates how to perform jamming detection in the application layer with the above-mentioned APIs.

3.1. Example Description

nw_demo.c, the demo file for jamming detection API, is in the *\components\quectel\ql_open\ql_examples\nw* directory of the module CSDK. It includes enabling jamming detection feature, modifying parameters, and the example of using the feature.

The entry function is *ql_nw_app_init()*, as shown below:

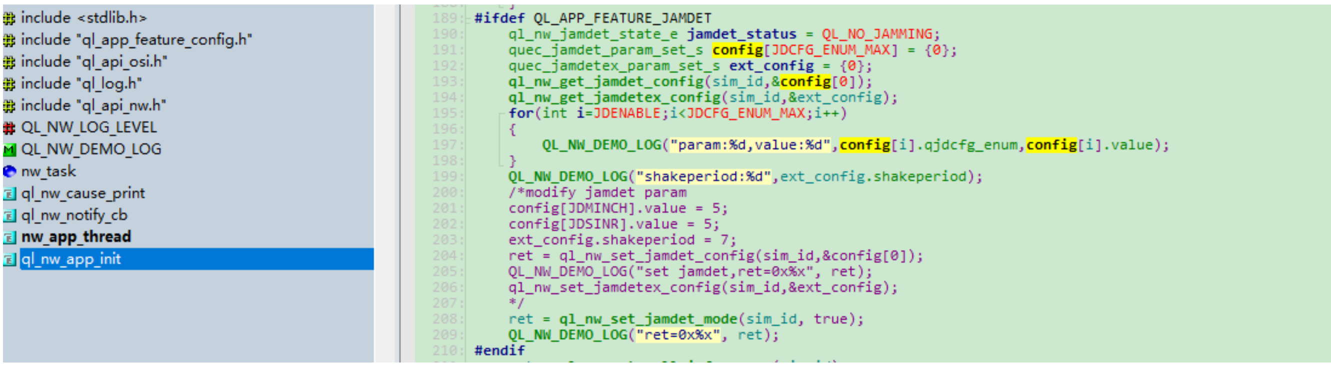

Figure 1: Entry Function

3.2. Get the Jamming Status

There are two ways to get the jamming status:

1. You can get the jamming status by calling *ql_nw_get_jamdet_status()*.
2. When the underlying layer detects a change in the jamming status, it actively reports to the App layer through the *QUEC_NW_JAMMING_DETECT_IND* event.

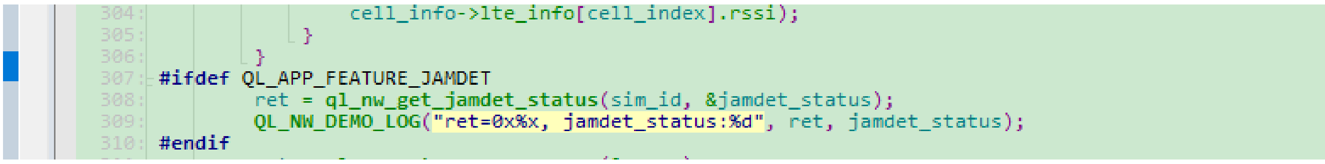

Figure 2: Actively Get the Jamming Status

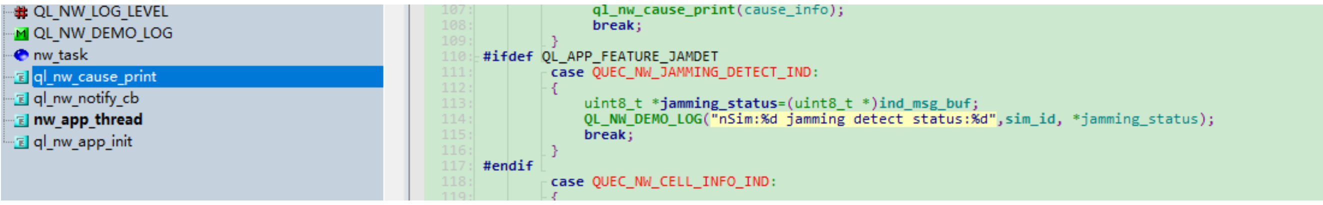

Figure 3: Event is Reported When Underlying Layer Detects a Jamming Status Change

4 Appendix References

Table 3: Related Document

Document Name
[1] Quectel_EC200U&EG91xU_Series_QuecOpen(SDK)_Quick_Start_Guide

Table 4: Terms and Abbreviations

Abbreviation	Description
API	Application Programming Interface
App	Application
GSM	Global System for Mobile Communications
ID	Identifier
IoT	Internet of Things
LTE	Long-Term Evolution
NW	Network
PLMN	Public Land Mobile Network
PS	Packet Switch
RRC	Radio Resource Control
RTOS	Real-Time Operating System
SDK	Software Development Kit
SIM	Subscriber Identity Module
URC	Unsolicited Result Code
(U)SIM	(Universal) Subscriber Identity Module